

PORTFOLIO DECISION SYSTEM (PDS)

Architecture, Governance, and Operating Framework

A framework for combining independent strategy targets into a single monitored portfolio

SYSTEM DOCUMENTATION v1.3

This document describes the three-layer PDS architecture - PDS Core, Reinforcement-Learning Adaptive Risk Control, and Dynamic FX - and the rules for decision timing, accounting, evidence, and public reporting.

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Document Summary

Field	Details
Document	Portfolio Decision System (PDS): Architecture, Governance, and Operating Framework
Version	1.3
Release date	2026-09-22
System type	Portfolio Decision & Operating System
Public scope	Validated canonical operating mirror of current Core, Adaptive, Preview, FX, Performance, and Portfolio information. Environment-specific infrastructure, brokerage-account data, and protected implementation details are excluded from publication.
Current architecture	Layer 1: PDS Core (F2R + ADAA). Layer 2: PDS Adaptive - Reinforcement-Learning Adaptive Risk Control. Layer 3: Dynamic FX - monthly USD/KRW implementation overlay.
Decision and rebalance cycle	Monthly signals; rebalance at the first U.S. trading-day close available to both strategies after the signal month ends.
Performance measurement	Daily returns are the performance source of record. Monthly and annual figures are derived from the continuous daily series, using 252 trading days for annualization.
Disclosure policy	Public and local dashboards carry the same validated operating information. Publication excludes local paths, runtimes, caches, credentials, repository/debug infrastructure, brokerage-account information, and protected controller artifacts.

How this document is organized

Sections 1-4 explain what PDS does and how the three operating layers fit together. Sections 5-9 cover construction, timing, performance, portfolio status, Dynamic FX, and governance. Sections 10-12 cover monitoring, operations, and public disclosure. Sections 13-15 cover release validation, related methods, and limitations. The appendices summarize decision status, public resources, and the v1.3 specification.

Terminology

In this document, F2R refers to Forecast-to-Rank Allocation and RL to Reinforcement Learning. Dynamic FX is the monthly USD/KRW overlay used for currency implementation, monitoring, and performance reporting in PDS.

OVERVIEW

1. Executive Overview

Portfolio Decision System (PDS) combines target portfolios from independently developed investment strategies into a single portfolio process. It leaves each strategy's investment logic unchanged and standardizes only the portfolio-level rules for integration, timing, accounting, monitoring, and disclosure. Source ownership and decision provenance remain traceable.

PDS has three operating layers. PDS Core combines Forecast-to-Rank Allocation (F2R) and ADAA under the current strategic allocation. PDS Adaptive starts from Core and adds Reinforcement-Learning Adaptive Risk Control, a bounded risk-control layer rather than a standalone alpha or security-selection model. Dynamic FX is a separate currency layer that applies the same USD/KRW rule, execution clock, and cost schedule to the relevant Core or Adaptive USD return path. Keeping the layers separate makes attribution and decision authority clear.



A useful distinction

F2R and ADAA source targets, the Core Official Decision, Adaptive decisions, Dynamic FX decisions, Forward Preview, Forward Shadow, and historical evidence do not carry the same authority. Core defines the base portfolio; Adaptive changes only its own risk budget; Dynamic FX changes only the currency layer.

What PDS does

- Combines strategy targets into one portfolio using a consistent set of portfolio rules.
- Applies the same rebalance, turnover, cost, and performance rules when the strategies are evaluated together.
- Keeps the current Official Decision separate from Forward Preview and Forward Shadow outputs.
- Retains the last valid Official Decision when required data or timing checks cannot be reconciled.
- Publishes the validated public view of Core, Adaptive, Preview, FX, Performance, and Portfolio information after release checks pass.
- Runs PDS Core and PDS Adaptive in parallel so Core remains unchanged while Adaptive risk control is evaluated and operated separately.

What PDS does not do

- It does not redesign the investment logic of F2R or ADAA.
- It does not treat Forward Preview, Forward Shadow, or FX research as an approved portfolio decision.
- An Official Decision records the PDS target; it is not a broker fill or an account-level trade confirmation.
- The public PDS dashboard is not a reduced investment view. It shows the same validated operating information as the local dashboard while excluding local paths, runtimes, caches, credentials, repository controls, and debug tooling.
- PDS Adaptive does not overwrite or backfill PDS Core history, and an Adaptive failure does not change Core portfolio authority.
- RL does not select securities or replace Core. It adjusts the Adaptive risk budget within the approved bounds.

ARCHITECTURE

2. System Architecture

PDS has three layers. Approved ADAA and F2R targets form Core. Adaptive applies a separately controlled Reinforcement-Learning risk-control layer to Core. Dynamic FX adds the monthly currency-implementation view for the Core and Adaptive USD paths. The layers share timing and accounting rules where necessary, but their decisions and historical records remain separate.

Stage	Step	What happens
01	Strategy targets	Each strategy produces a target portfolio according to its own rules.
02	Common format	Targets are put into a consistent asset and weight format so they can be combined without changing the underlying strategy rules.
03	Input checks	Strategy identity, date, timing, completeness, and weight consistency are checked before use.
04	Strategy review	Research evidence and practical portfolio value are reviewed before a strategy is approved for a specific role.
05	Portfolio combination	Approved targets are combined using the current PDS allocation policy.
06	Portfolio decision	The monthly PDS Core target is set under the common decision/execution clock. PDS Adaptive may apply its own risk budget, and Dynamic FX uses a separate monthly currency decision without changing the underlying USD target.
07	Monitoring and review	Core, Adaptive, Dynamic FX, Current Mark, Forward Preview, Forward Shadow, and realized performance are monitored under explicit clocks and evidence classes, with each view retaining its own status.

What PDS standardizes

PDS never reconstructs a missing strategy output. It uses only source targets that can be traced to the underlying strategy and that pass the required date, timing, and completeness checks. If a required input cannot be reconciled, the update is not adopted and the last valid Official Decision remains in place.

Independent strategies, common portfolio rules

F2R and ADAA retain their own investment logic. PDS standardizes only the elements that must be common at portfolio level: target format, timing, portfolio combination, performance measurement, and monitoring.

How the components fit together

System	Role in PDS	What remains separate
F2R	Produces a machine-learning cross-asset target portfolio used by PDS.	F2R retains its forecasting, ranking, and target-construction rules.
ADAA	Produces a multi-asset target portfolio used by PDS.	ADAA retains its allocation rules and target construction.
PDS	Combines the strategy targets and applies the common portfolio rules.	PDS does not recreate or override the underlying strategy models.
PDS Adaptive	Applies Reinforcement-Learning Adaptive Risk Control as a bounded layer on top of PDS Core.	It is not a standalone alpha or full portfolio-selection engine. The current state may be public, while exact controller parameters and training artifacts remain internal.
Dynamic FX	Applies the monthly USD/KRW overlay to Core or Adaptive USD paths.	The currency layer is separate from the underlying portfolio decision; spot-based monitoring is not broker-level forward/NDF execution evidence.

STRATEGY INPUTS

3. Strategy Inputs and Responsibilities

F2R and ADAA remain responsible for their own calculations. PDS takes their dated target portfolios, verifies the strategy identity, dates, timing, and weights, and combines them under PDS rules. A missing strategy output is never recreated inside PDS.

Checks before a target is used

- The correct strategy, target date, and relevant version are identified.
- Signal month, holding month, market-data cutoff, and rebalance timing line up.
- Weights are complete, properly scaled, and traceable to the strategy output.
- Strategy-level costs and PDS rebalancing costs are kept separate so the same cost is not counted twice.
- Missing, stale, or conflicting outputs are not replaced by estimates; the last valid Official Decision remains in place.

Research history and PDS operating history

A strategy's original research record may use execution or cost assumptions that differ from PDS. PDS preserves that record unchanged. For cross-strategy evaluation, PDS builds a separate operating history under common timing and accounting rules. Adaptive historical evidence is a selection-aware BACKTESTED_SANDBOX reconstruction from the approved frozen policy; it is not the operational decision ledger. Operational Adaptive history starts only after deployment and valid execution, and research history is never spliced into it.

Record	Purpose	How it is used
Original strategy research	Preserves the strategy's original backtest, timing, and cost assumptions.	Used to interpret claims about the strategy itself.

Record	Purpose	How it is used
PDS operating history	Measures the approved strategies under the same PDS timing and accounting rules.	Used to evaluate the combined portfolio under the PDS timing and accounting rules.
Public operating mirror	Publishes the validated current operating information after sanitization and release checks.	Used to communicate current monitored state and completed evidence without exposing environment-specific infrastructure.
Adaptive operational history	Post-deployment decisions and realized accounting only.	Separate from frozen-policy backtest evidence and PDS Core; no retrospective splice or backfill.

Historical and operational records are kept separate
Original strategy research, PDS Core operating history, Adaptive frozen-policy evidence, Adaptive operational decisions, and the public operating mirror remain separately identifiable.
Publication does not change their evidence class or decision authority.

CORE PORTFOLIO

4. Current Core Construction

PDS Core is the base portfolio. Its current monthly top-level target is F2R 25% and ADAA 75%. Each strategy determines its own asset weights; PDS combines the source targets under the common monthly execution, turnover, cost, and daily-accounting rules. The SlackQuant landing page focuses on the architecture rather than the exact ratio; the dashboard and this document provide the construction detail.

PDS Core monthly target = F2R 25% + ADAA 75%

Total target weight = 100%

The ratio is part of the detailed system construction; the SlackQuant landing page does not present it as the identity of the system.

The provider allocation is set at each monthly rebalance, not reset every day. After rebalancing, both provider weights and underlying asset weights drift with market returns until the next scheduled rebalance.

PDS Adaptive

PDS Adaptive is a parallel Core-based variant driven by an approved frozen RL policy. In NORMAL state, its target risk budget is 100% of Core. In DEFENSIVE state, it can scale down Core risk exposure and allocate the remainder to BIL. The historical Adaptive series is reconstructed from the frozen policy over the certified out-of-sample period and remains selection-aware evidence; the current portfolio table comes from a separate operational decision ledger. The public dashboard may show the current state and risk budget, while training artifacts, model seeds, thresholds, and exact controller parameters remain internal.

Target and portfolio weights

Measure	Meaning	Use
Official target	Portfolio weights set for the next holding period at the monthly rebalance.	Reference for the scheduled rebalance.
Post-rebalance weights	Portfolio weights immediately after the scheduled rebalance.	Starting weights for the next holding period.
Pre-rebalance weights	Portfolio weights immediately before the next rebalance after market movements.	Used to calculate turnover and rebalancing costs.
Current Mark	Provisional valuation of the incomplete holding period.	Monitoring only; does not change completed history or the Official Decision.
Execution-day opening exposure	The drifted portfolio held from the prior close into the scheduled execution close.	Earns the execution-day return before the new target is applied; this can differ between Core and Adaptive even when the new close target is identical.

Why weight drift matters

Resetting provider weights to their strategic targets every day would define a different strategy, with different exposures, turnover, and returns. PDS therefore lets weights drift between monthly rebalances and restores the target allocation only at the next scheduled rebalance.

Key point

The provider allocation is set at the monthly rebalance and then allowed to drift; it is not reset to constant weights each day.

TIMING

5. Decision and Rebalance Timing

PDS uses a monthly decision cycle. After the signal month closes, the next target is determined from information available at that cutoff. Rebalancing occurs at the first U.S. trading-day close available to both F2R and ADAA. The inherited portfolio earns the close-to-close return through that execution date; transaction cost is booked at the close; the new target starts with the next return interval. The same rule applies to Core and Adaptive. As a result, an Adaptive risk budget of 100% can still produce a different execution-day return if Adaptive entered the day with a different inherited book.



How the monthly cycle works

- The completed signal month determines the target for the next holding period.
- The portfolio is not rebalanced at the same month-end close that produced the signal.
- The existing portfolio remains in place through the rebalance close.

- The new weights apply from the next close-to-close return interval.
- Monthly performance is labeled by the holding month in which the return is realized.
- Calendar-year performance is assigned to the year in which the holding-period returns occur.

When dates do not line up

Strategy targets are combined only when their signal dates and relevant market-data dates are aligned. If a required strategy output is missing or not yet available, PDS does not estimate it. The last valid Official Decision remains in force until the inputs can be reconciled.

Why timing matters

The signal cutoff, execution close, and first return interval of the new target are different points in time. A 100% Adaptive risk budget set at month-end means the target will match Core after execution; it does not replace the inherited Adaptive book that earns the execution-day return. This is part of the portfolio definition, not a reporting convention.

PERFORMANCE MEASUREMENT

6. Performance Measurement and Transaction Costs

All PDS performance statistics are derived from continuous daily returns. Monthly, year-to-date, annual, cumulative, drawdown, volatility, Sharpe, and Calmar figures use this daily series, with 252 trading days for annualization. Core and Adaptive follow the same convention, but they can differ on an execution day because that day's return is earned by the pre-trade book before the new close target is applied.

$$\begin{aligned}\text{Daily wealth: } W_d &= W_{(d-1)} \times (1 + r_{d,\text{net}}) \\ \text{Monthly return: } R_{\text{month}} &= W_{\text{month-end}} / W_{\text{prior-month-end}} - 1\end{aligned}$$

Return and risk calculations

Daily returns are the source frequency. Annualized volatility and Sharpe use 252 trading days, and drawdown is measured from the running peak of daily wealth.

Turnover and rebalancing cost

Turnover is calculated from the portfolio weights immediately before the rebalance and the new target weights. PDS defines one-way turnover as one half of the sum of the absolute weight changes.

$$\begin{aligned}\text{One-way turnover} &= 0.5 \times \sum_i | \text{target weight}_i - \text{pre-rebalance weight}_i | \\ \text{PDS rebalancing cost} &= 2 \times \text{one-way turnover} \times 0.001\end{aligned}$$

The current cost assumption is 10 basis points per side. A full portfolio replacement (100% one-way turnover) therefore carries a 20-basis-point round-trip cost.

Avoiding double counting

Costs already reflected in the F2R or ADAA return series are not charged again at PDS level. PDS adds only the incremental cost of moving from the drifted pre-trade book to the new PDS target. Adaptive follows the same rule and rebalances directly from its own inherited book; it does not first pay for a hypothetical Core rebalance. On an execution date, the old book earns that day's return, turnover and cost are booked at the close, and the new target applies from the next return interval.

Consistency check

All published PDS return figures must follow the same timing, turnover, cost, and daily performance rules. If two views do not reconcile, the difference is investigated before release.

PORTFOLIO STATUS

7. Official Decision and Monitoring Views

The dashboard distinguishes portfolio decisions from monitoring views. The Core Official Decision is the current Core target, while PDS Adaptive runs in parallel with its own state and risk budget. Current Mark, Forward Preview, and Forward Shadow remain monitoring views. The public dashboard can show current Core and Adaptive state, portfolio detail, Preview, performance, and Dynamic FX information under the same timing rules as the local dashboard.

View	What it shows	How it is used
Core Official Decision	The target portfolio in effect for the current holding period.	Used as the PDS portfolio target. It is not a broker fill or trade confirmation.
Current Mark	A provisional view of performance for the current, incomplete holding period.	Monitoring only; it does not change the Official Decision or completed performance history.
Forward Preview	What the next target would look like if the current month ended using the latest available data.	Monitoring only; it is not executed and no Forward Preview performance is reported.
Forward Shadow	A separately tracked alternative to the Active Core used for comparison.	Not traded as part of the Active Core; any performance is kept separate.
Public Operating Dashboard	Validated public copy of the current operating state.	Shows current monitored state and completed evidence under the same clocks as the local dashboard.
PDS Adaptive	Parallel Core-based variant with NORMAL or DEFENSIVE risk-control state.	Current state, risk budget, holdings, Preview, and performance may be shown publicly; low-level controller internals remain protected.

More recent does not mean official

A newer Forward Preview or Forward Shadow result does not replace the Official Decision. It simply adds more recent monitoring information.

Changing portfolio authority

A monitoring view becomes part of Core only through the regular monthly decision process. PDS Adaptive changes only its own parallel portfolio through the risk-control layer. A newer timestamp does not change either status.

KRW INVESTOR CONTEXT

8. Currency Exposure and the FX Overlay

For a KRW-based investor holding U.S.-listed multi-asset ETFs, exchange-rate moves can materially affect portfolio returns and risk. PDS therefore manages currency exposure at the total-portfolio level rather than embedding a separate FX decision inside one underlying strategy.

FX implementation and clocks

FX view	Status	How to interpret it
Dynamic FX operating view	Current monitored implementation layer	Shows the current Official and Preview monthly hedge state, Current MTD, and completed Core/Adaptive Dynamic-FX performance under the defined spot-sensitivity rule and cost schedule.
Broker-level forward/NDF realization	Not certified by the current spot proxy	Realized hedge P&L would require tradable forward/NDF data, actual availability, carry/basis, contract-roll rules, execution, and realized transaction costs.
FX clock terminology	Three distinct timestamps	Latest raw FX observation = newest source-feed timestamp; Latest admissible FX observation = newest FX point allowed by the PDS/common calendar and usable for z-score/hedge decisions; Portfolio/Preview market-through = latest date through which the relevant portfolio/Preview market inputs are complete. These dates need not be identical.

Dynamic FX is the monthly USD/KRW implementation layer for both PDS Core and PDS Adaptive. The same FX rule, execution clock, and cost schedule apply to both; only the underlying USD portfolio path may differ. The public dashboard can show the current Official FX decision, Preview state, Current MTD, and completed performance. These spot-based results are an implementation proxy, not realized broker-level forward/NDF hedge P&L.

Why the distinction matters

- Spot-FX sensitivity is not the same as realized forward/NDF hedging performance.
- Forward carry, basis, contract rolls, and changes in hedge ratios can affect actual hedging results.
- An FX series developed in a strategy research project is not automatically adopted by PDS.
- A historical FX series should be interpreted using its stated signal, admissibility, and market-through clocks rather than treated as a substitute for an executed hedge record.

GOVERNANCE

9. Research Evidence, Portfolio Usefulness, and Adoption

PDS separates three questions: Does the evidence support the research claim? Does the strategy add value in the portfolio role being tested? Has it actually been adopted into the Official portfolio? A positive answer to one does not imply a positive answer to the others.

Review	Question	What it determines
Research review (Gate R)	Does the evidence support the research claim being made?	Whether the research claim passes, fails, or remains inconclusive.
Portfolio-role review (Gate P)	Does the strategy add enough practical value for the portfolio role being tested?	Whether the strategy is suitable for that role, unsuitable, or still inconclusive.
Official adoption	Has the strategy or allocation actually been approved for the PDS portfolio?	Only an explicit adoption decision changes the Official portfolio.

These are separate decisions

A strategy can have sound research evidence without being useful for the role being considered. It can also look useful in a portfolio test without supporting a new research claim. Neither result changes the Official Core unless the strategy is explicitly adopted.

When a review does not pass

If required data, timing, or evidence cannot be verified, the proposed update is not used. A Forward Preview issue does not alter the last valid Core Official Decision. A strategy stays outside the Official Core until a later review supports adoption. Adaptive follows the same fail-closed principle: if the frozen policy, state inputs, or integrity checks fail validation, it produces no new decision, while Core continues independently.

MONITORING

10. Monitoring Views

PDS monitors the current Core Official portfolio, PDS Adaptive, Forward Preview, Forward Shadow, Current Mark, Dynamic FX, and realized performance. Each view has a defined role and clock so monitoring information is not mistaken for a portfolio decision. After validation, the public dashboard shows the same operating information while excluding environment-specific infrastructure and protected implementation details.

Dashboard views

Dashboard view	What it shows	How to interpret it
PM Cockpit	The current Core Official portfolio decision and the main changes from the previous decision.	Official portfolio view.
Performance	Core, Core + Dynamic FX, Adaptive, and Adaptive + Dynamic FX performance with explicit support windows.	Completed history remains separate from provisional Current MTD; Adaptive frozen-policy history retains its own evidence class.
Portfolio Weights	Current strategy targets and the resulting asset-level PDS target.	Target weights are distinguished from weights that have drifted since the last rebalance.
Forward Shadow	Performance and allocations for a separately tracked alternative to the Active Core.	Comparison only; it does not become the Active Core because it is displayed or performs well.
Forward Preview	The target that would be indicated if the current month ended using the latest available data.	Monitoring only; it is not executed and no Forward Preview performance is reported.
FX	Current Official and Preview FX state, admissible observation clock, Current MTD, hedge history, and completed performance.	The same FX rule and cost schedule apply to Core and Adaptive USD paths; spot-based results are not broker-level NDF execution evidence.
PDS Adaptive	Current Adaptive state, risk budget, holdings, Preview, and performance alongside Core.	Parallel to Core; frozen-policy historical evidence and post-deployment operational decisions remain separate authorities.

When data are unavailable

If a strategy update, Forward Preview input, timing reference, or Current Mark is unavailable, PDS keeps the last valid Official information and marks the newer item as unavailable. Missing data are not estimated merely to keep the dashboard populated.

OPERATIONS AND PUBLICATION

11. Operating and Publication Process

PDS decisions are produced in the controlled operating environment. Publication creates a sanitized copy of the validated dashboard rather than a separate delayed or simplified investment product. It does not create new trading authority; it packages and distributes information that has already passed the operating checks.



Main operating steps

Step	Purpose
Refresh and rebuild	Update strategy inputs and rebuild the PDS dashboard. If a monitoring input is unavailable, the last valid Official Decision remains unchanged.
Review the current state	Check the Official Decision, Current Mark, Forward Preview, Forward Shadow, and related changes before any release is prepared.
Prepare the public update	Freeze and sanitize the dashboard, remove environment-specific infrastructure, and verify that the public version preserves the validated operating information and clocks.
Approve and publish	Review the changes, confirm the published figures, and release the update to the SlackQuant site.

Publication checks

1. Check the PDS public figures, labels, links, and disclosure boundaries.
2. Build the public pages and inspect the rendered result.
3. Confirm that the downloadable and linked resources match the approved release.
4. For a new system-document version, review the rendered document page-by-page; fix and re-render each page with a finding before moving to the next page.
5. Publish only after the checks above pass.

Final review remains explicit

Automation can update data, run checks, and prepare a release, but a substantive public update still receives a human review before publication.

PUBLIC DISCLOSURE

12. Public Disclosure

The public PDS dashboard is the validated canonical operating mirror of the local reader-facing dashboard. Current Core and Adaptive state, portfolio detail, Preview, Dynamic FX monitoring, and performance may be published. The public/private boundary is mainly technical: local paths, runtimes, caches, credentials, repository controls, debug tooling, brokerage-account information, and protected model artifacts remain private.

Published / mirrored	Internal / suppressed
System architecture, provider relationships, governance, and system documentation	Local filesystem paths, runtime/cache/repository locations, credentials/secrets, and debug/runner tooling
Current Core and Adaptive decision state, detailed portfolio targets, and operating clocks	Low-level Adaptive controller training artifacts, model seeds, exact thresholds/parameters, and implementation-specific internal identifiers
Current Mark, Forward Preview, Forward Shadow where shown on the dashboard, and performance monitoring	Brokerage-account holdings, realized account P&L, order-routing infrastructure, and trade confirmations
Current Dynamic FX monitoring, Preview FX state, Current MTD, hedge history, and completed Dynamic-FX performance	Unpromoted research branches and non-canonical experimental artifacts
Research, Quantitative Methods, release validation, and versioned system documentation	Machine-specific provenance that is not needed to interpret the published system
Core, Core + Dynamic FX, Adaptive, and Adaptive + Dynamic FX performance with explicit support windows and evidence classes	Protected security-sensitive implementation details not required to reproduce the published operating state

How public performance is dated

Monthly performance is labeled by the month in which the portfolio return is realized, not by the month in which the signal was formed. Annual figures follow the same holding-period convention.

Status of public materials

Public PDS pages are research and monitoring tools, not execution interfaces or trade confirmations. Their dates and status follow the same decision, admissibility, and market-through rules used by the operating dashboard.

Publication rule

A release is blocked if figures, clocks, labels, portfolio detail, or cross-system identities do not reconcile, or if environment-specific or security-sensitive material appears in the public artifact. Publication does not create new portfolio authority.

RELEASE CHECKS

13. Release Validation

For v1.3, the Core/Adaptive architecture, execution-close accounting, Dynamic FX clock terminology, frozen-policy evidence treatment, public disclosure rules, performance history, and publication wiring were checked against the operating records. The table summarizes the release checks; it documents the system as released and is not a forecast.

Check	Result
Core allocation	The current monthly top-level Core target was confirmed as F2R 25% / ADAA 75%. The ratio is shown in the detailed documentation and dashboard rather than emphasized on the platform landing page.
Historical Core	The displayed history applies the current Core policy consistently throughout; earlier allocation policies are not spliced into the series.
Rebalance timing	The monthly decision and rebalance dates matched the timing rules described in Section 5.
Transaction costs	Drift-based turnover and the 10 bp-per-side cost assumption matched the operating calculation.
Performance measurement	The daily return series, 252-trading-day annualization, and drawdown calculations reconciled.
Data coverage at release	PDS Core common daily history ran from 1 May 2017 through 31 August 2026, covering 2,347 observations. Core-versus-Adaptive historical comparison uses common support from 3 May 2021 through 31 August 2026.
Current Mark	The Current Mark remained a monitoring view and did not alter the Official Decision or completed history.
Display consistency	Core, Adaptive, Preview, Dynamic FX, Performance, and Portfolio views matched the current terminology and operating clocks.
Frozen Adaptive inference	The approved frozen inference bundle passed historical policy parity, decision-boundary parity, reload parity, and operating-path parity before operational export was permitted.
Core non-mutation / failure isolation	PDS Core allocation and accounting remain unchanged. Adaptive failures are contained within the Adaptive branch and do not block or rewrite Core operation.
2026-08 Core vs Adaptive return reconciliation	PASS. The 2026-07 Adaptive decision was NORMAL with 100% risk budget and executed at the 2026-08-03 close. The 2026-08-03 return was earned by the inherited pre-trade books: Adaptive entered with about 54.14% risky exposure and then rebalanced to the same 25% F2R / 75% ADAA target as Core. From 2026-08-04 through 2026-08-31 the Core and Adaptive USD daily returns were identical. The resulting Dynamic-FX monthly returns reconcile to 3.4807315% for Core and 2.9984257% for Adaptive.

RELATED METHODS

14. Related Quantitative Methods

The SlackQuant Quantitative Methods library provides background on the techniques referenced in this document. The entries below are the methods most directly related to PDS.

ID	Method	Relevance to PDS
QM007	Portfolio Backtesting and Rebalancing	Explains monthly rebalancing, weight drift, and next-period return timing.
QM008	Drawdown, Underwater Duration, and Recovery	Covers drawdown depth, duration, recovery, and running peaks.
QM009	Turnover, Transaction Costs, and Net Performance	Explains turnover, transaction costs, and net portfolio returns.
QM011	Strategy Similarity and Decision Diversification	Provides the basis for comparing strategy decisions rather than return correlation alone.
QM013	Benchmark-Relative Portfolio Evaluation	Covers active return, tracking error, information ratio, relative wealth, and relative drawdown.
QM014	Information Timing, Data Vintages, and Real-Time Macro Data	Explains release timing, data vintages, revisions, and the information available at each decision date.
QM028	Reinforcement Learning for Adaptive Portfolio Risk Control	General background on RL-based risk control. PDS-specific frozen-policy binding, evidence classes, execution clock, and accounting are defined in this document.
QM029	Dynamic Currency Hedging and FX Overlay Accounting	General background on FX overlays. PDS-specific raw/admissible observation rules, market-through clock, monthly hedge decision, and Core/Adaptive treatment are defined here.

Methods library: research.slackquant.com/methods/

How these references are used

The Methods library explains the general techniques; this document specifies how PDS uses them. QM028 covers Reinforcement Learning for Adaptive Portfolio Risk Control, while this document defines the PDS frozen-policy binding, clocks, evidence classes, and accounting. QM029 covers Dynamic Currency Hedging and FX Overlay Accounting, while this document defines the PDS FX observation rules, monthly decision clock, and Core/Adaptive overlay treatment.

LIMITATIONS

15. Limitations and Revision Policy

Limitations

- Historical results, including observations from live operation, do not guarantee future returns.
- The current F2R/ADAA allocation is the present PDS Core policy. It is a system design choice, not a claim that the ratio is universally or permanently optimal.
- F2R and ADAA can still share market exposures even though their decision processes differ. Their combination should therefore be evaluated from the observed portfolio results rather than assumed to be diversified by design.
- Transaction costs are modeled assumptions. Actual trading costs can differ.
- Dynamic FX uses spot-FX sensitivity and an assumed hedge-renewal cost schedule. It should not be read as realized broker-level forward/NDF hedging performance.
- Forward Preview and Forward Shadow are monitoring tools and do not change the Official portfolio unless they are separately adopted through the regular decision process.
- Adaptive backtest evidence was selected during exploratory research and is not treated as independent confirmatory evidence. Operational Adaptive history is tracked separately.
- This document describes the PDS portfolio framework. Broker fills, custody records, account holdings, and trade confirmations are outside its scope.

When a new version is needed

Change	Update needed
Presentation-only edit	Normally no new version unless the meaning changes.
Routine data update or new completed month	No new version; the historical record is simply extended.
Core strategy membership or target allocation	New document version.
Rebalance timing, turnover, or transaction-cost treatment	New document version.
Definition or role of Official Decision, Forward Preview, or Forward Shadow	New document version.
Public disclosure policy	New document version.
Dynamic FX role, implementation authority, or accounting treatment	New document version.
PDS Adaptive role, policy binding, evidence boundary, execution semantics, or performance-accounting treatment	New document version.

Versioning rule

A new version is issued when portfolio construction, timing, cost treatment, decision status, or disclosure policy changes materially. Routine monthly data updates do not require a new version.

Appendix A. Decision and Publication Status

View	Current?	Can change Core target?	Performance?	Published?
Core Official Decision	Yes	Yes - through the regular monthly decision process	Realized under the execution-close accounting rule	Yes - current target and operating clocks are shown in the public dashboard
Current Mark	Yes - incomplete period	No	Provisional current-period view	Yes - monitoring status is explicit
Forward Preview	Yes - incomplete period	No	No executed Preview performance	Yes - target-only/non-authoritative status is explicit
Forward Shadow	Yes - monitoring	No	Separate comparison performance may be tracked	Yes where shown on the public dashboard; display alone does not change its status
Public Operating Mirror	Current validated snapshot	No - publication adds no portfolio authority	Mirrors current monitored and completed evidence	Yes
Dynamic FX	Yes - separate monthly currency layer	No - does not change the underlying Core target	Current MTD and completed overlay performance	Yes - current Official/Preview state and completed evidence; broker-level NDF execution is not implied
PDS Adaptive	Yes - parallel operating variant	Changes only the Adaptive target through its risk-control layer; never changes Core	Frozen-policy historical evidence plus separate post-deployment operational accounting	Yes - current state, risk budget, and portfolio can be shown; low-level controller internals remain protected

Reading the table

- The Core Official Decision changes the Core target through the regular monthly process. PDS Adaptive changes only its own parallel target through the risk-control layer. Forward Preview, Forward Shadow, and Current Mark remain monitoring views. Publication does not turn a monitoring view into an Official Decision.

Appendix B. PDS Resources

Resource	Purpose	Location
PDS System page	Overview of the system, PDS Core / PDS Adaptive architecture, current Core providers, and related public resources.	research.slackquant.com/systems/pds/
PDS Dashboard	Validated public operating view for Core, Adaptive, Preview, FX, Performance, and Portfolio.	research.slackquant.com/systems/pds/dashboard/
PDS System Documentation	Detailed explanation of Core/Adaptive construction, timing, accounting, monitoring, governance, and disclosure.	Linked from the PDS System page
Quantitative Methods	Background on the quantitative methods referenced in this document.	research.slackquant.com/methods/
Research	Separate empirical papers and research records related to the strategies and methods.	research.slackquant.com/research/

Start with the System page for an overview. The Dashboard shows the validated public operating view for current Core, Adaptive, Preview, FX, Performance, and Portfolio information. This document and the Methods library provide the supporting system and methodology detail.

Appendix C. PDS v1.3 Summary

Item	PDS v1.3
System	Portfolio Decision System (PDS)
System type	Portfolio Decision & Operating System
Current Core providers	F2R and ADAA
PDS Core	Base portfolio with a monthly top-level target of F2R 25% + ADAA 75%. The platform landing page emphasizes the architecture rather than the ratio.
Rebalance frequency	Monthly
Between rebalances	Provider and asset weights drift with market returns; there is no daily reset to strategic provider targets.
Rebalance timing	First common U.S. trading-day close after the signal month. The inherited pre-trade book earns through that close; the new target starts from the next close-to-close interval.
Turnover measure	One-way turnover is half the sum of the absolute changes between pre-rebalance and target weights.
Transaction-cost assumption	10 basis points per side.
Performance calculation	One continuous daily return series; monthly and annual results are derived from it.
Forward Preview	Monitoring only; not executed and no Forward Preview performance is reported.
Forward Shadow	Separate comparison to the Active Core; not part of the Official portfolio unless adopted.
Public disclosure	Validated canonical operating mirror. Current operating information is public; environment/security infrastructure and protected implementation details remain private.
Dynamic FX	Third operating layer: monthly USD/KRW overlay applied under the same rule, execution clock, and cost schedule to Core and Adaptive USD paths. Current Official/Preview state and completed performance may be displayed.
FX evidence boundary	Spot-based operating sensitivity with assumed renewal cost is not broker-level forward/NDF realized P&L. Raw FX observation, admissible FX observation, and portfolio/Preview market-through are distinct clocks.
PDS Adaptive	Parallel Core-based variant using Reinforcement-Learning Adaptive Risk Control; RL is bounded risk control, not a standalone alpha or security-selection engine.
Adaptive operating states	NORMAL or DEFENSIVE. Current state and risk budget may be shown in the public dashboard; exact controller recipe, seeds, thresholds, and training artifacts remain internal.

Item	PDS v1.3
Adaptive evidence boundary	Historical series: selection-aware BACKTESTED_SANDBOX reconstruction from the approved frozen policy. Current and operational decisions: separate persistent operational ledger. No splice or retrospective backfill.
Adaptive failure behavior	Fails closed within the Adaptive branch; PDS Core remains independent and continues to refresh.

About this version

Version 1.3 describes the PDS framework in effect on 22 September 2026. It updates the documentation for the three-layer Core / Adaptive / Dynamic FX architecture, the role of Reinforcement-Learning Adaptive Risk Control, frozen-policy evidence treatment, execution-close accounting, FX clock terminology, and the public dashboard contract. The 2026-08 Core-versus-Adaptive return difference was reconciled to the inherited execution-day book and transaction costs; no performance-series correction was required.

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System documentation only. Not investment advice.